

KI-VQA benchmarks normally only evaluate end-task accuracy...

Instead, we enable evaluation of the entire KI-VQA pipeline, consisting of the following stages:

For stage-based analysis we release CRAG-MM-Diagnostics, a benchmark with a range of metadata, such as:

1

Language-based visual grounding

Do models know which part of the image this linguistic expression denotes?

We enable stage-based evaluation through ROI accuracy, and analyses across linguistic and visual complexity. We examine the effect of linguistic ambiguity by disambiguating ambiguous questions.



target ROIs

"Is the grey car sold in South America?"

disambiguated linguistic expressions

2

Object identification

Do models know which target object this is?

We evaluate entity recognition accuracy, decoupling it from stage 1 by providing ROIs explicitly.



target entity name/URL

"Lexus GX 470"

3

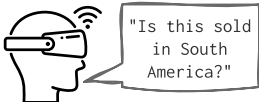
Knowledge retrieval & reasoning

Can models answer knowledge-intensive questions?

We evaluate this decoupled from stages 1 and 2 via verbose textual-only questions.

"Is the Lexus GX 470 sold in South America?"

text-only ablation



"No, the Lexus GX 470 is sold in North American and Eurasian markets."

